

Speech Tech

Week-4

Vector Quantizaⁿ

- we were talking about vector quantizaⁿ. It is technique to compress the info. $\{V\}$ large vector becomes $\{G\}$ smaller group of vectors. It happens that each vector in $\{G\}$ is the centroid of group of vectors it represents.
- practical ex. related to speech tech is its applicaⁿ in linear predictions.
- but its still there are some issues in creating excitaⁿ $\rightarrow$ but it has some computer-synthetic voice in the actual voice input.
- There ^{are} 2 common methods for vector quant.
 - 1. K means
 - 2. Linde-Buzo-Gray (LBG) algorithm
- K means
 - $\rightarrow$ we assume the no. of clusters
 - $\rightarrow$ first we randomly some set of pts. as cluster centroids

- based on 3 centroids, we create 3 clusters by associating every vector with nearest centroid (use - Euclidean distance).
- we reestimate the centroids using the vectors that are associated with each.
- we iterate until convergence is reached.

- LBG

- take 1st step - by calculating the 1 centroid for all vectors
- given centroid c , we defined $(c + \delta)$ & $(c - \delta)$, where δ is small distortion. We assign each vector to nearest clusters.
- repeat this until computed distortion after splitting the centroids becomes greater than before splitting the centroids.

L2 Markov Model

- we have so far learned to model non-sequential data.

- But sound / speech is sequential data where order is impt. Eg. DNA seq., text document, stock market, weather.
- Markov Chain - Weather eg.
 - 3 possibility → sunny, cloudy, rainy
 - the seq. can be pictured as a state machine where each state represents a distinct weather.
 - The arrows from 1 state to another represent state transitions & are associated with state transition possibilities.
 - today's weather can be solely decided by yesterday's weather in 1st order Markov chain.
- one method, based on 100 yrs data, how many consecutive sunny days were present? This can tell if today is sunny, what is the chance that tomorrow also will be a sunny day.

$$\frac{N(\text{sun, sun})}{N(\text{sun})}$$

- we can create a matrix $\begin{matrix} & \begin{matrix} C & S & R \end{matrix} \\ \begin{matrix} C \\ S \\ R \end{matrix} & \begin{bmatrix} & & \\ & & \\ & & \end{bmatrix} \end{matrix}$

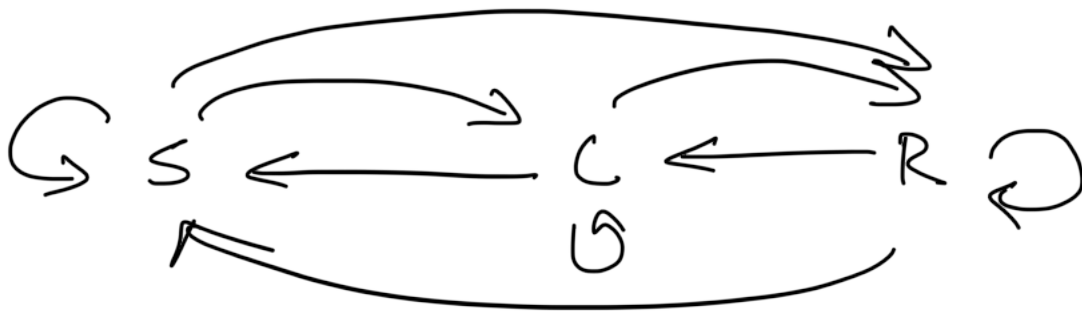
obtained from historical data ←

→ supervised machine data

- for a pattern like
 $R \ C \ C \ S \ S \Rightarrow P(R/S) \ P(C/R) \ P(C/C) \ P(S/C) \ P(S/C)$

so we take that seq. which provides the max. probab. for the 5-day forecast.

L3 Markov Chain - Eq.



weather eq. (from previous lecture)

- continued explainⁿ from the supervised learning $\rightarrow$ probability.
- We have to compute N^2 transiⁿ possibilities where there are N states. Each probability corresponds to a state transition. Once we obtain the transiⁿ probabi. matrix, we can solve various problems in markov chain.
- we can calculate the probability of a weather sequence using the multiplicⁿ rule.

L4 Predicting Weather Sequence

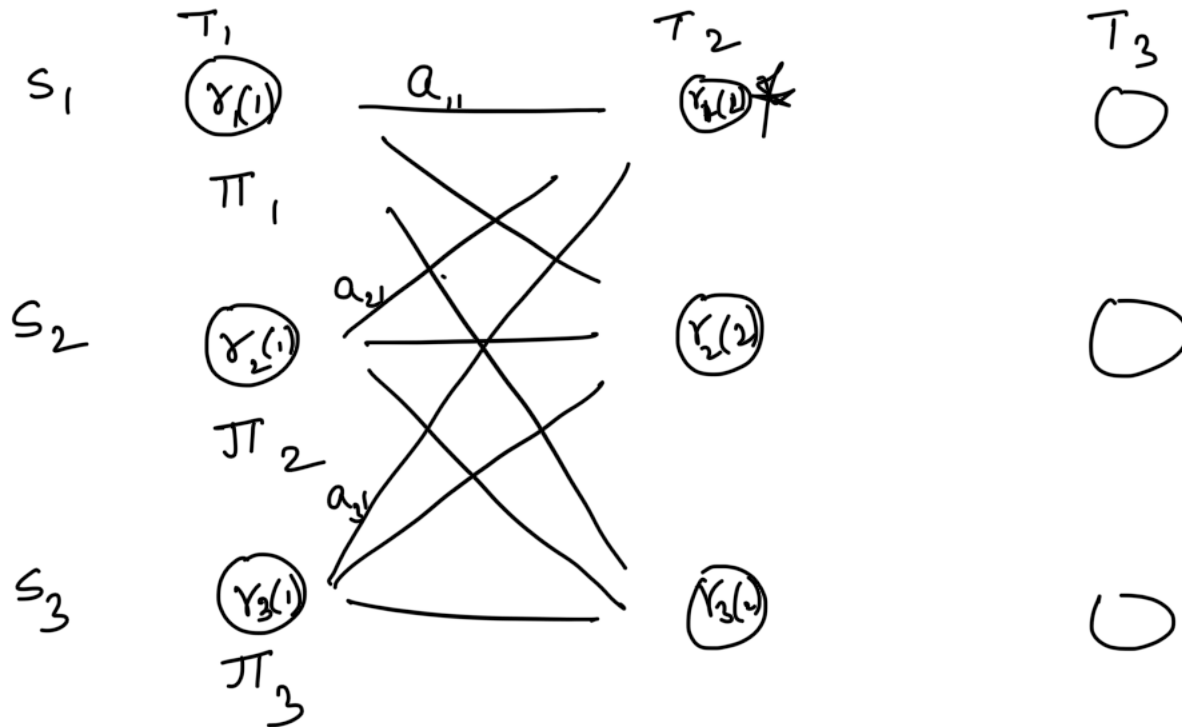
- the current method that we are employing is called a greedy approach.
 $s \rightarrow s \rightarrow s \rightarrow s$ (because always on the same row).
- greedy does not give best results.
To get the best compute probability of all sequences.
- for 3 days $s \rightarrow 3 \times 3 \times 3 = 27$ paths
5 days $\rightarrow 3 \times 3 \times 3 \times 3 \times 3 = 243$

$$\boxed{\begin{matrix} \text{no. of days} \\ \text{States} \end{matrix}} \Rightarrow \boxed{N^T}$$

N states
 T len of seq.

- This is very expensive. Viterbi Algo. $\rightarrow$ finds the most probable paths in very less time complexity.

L5 Viterbi Algo - Intro



$$\gamma_1(2) = \max \left\{ \begin{aligned} &\gamma_1(1) \cdot a_{11} \\ &\gamma_2(1) \cdot a_{21} \\ &\gamma_3(1) \cdot a_{31} \end{aligned} \right\}$$

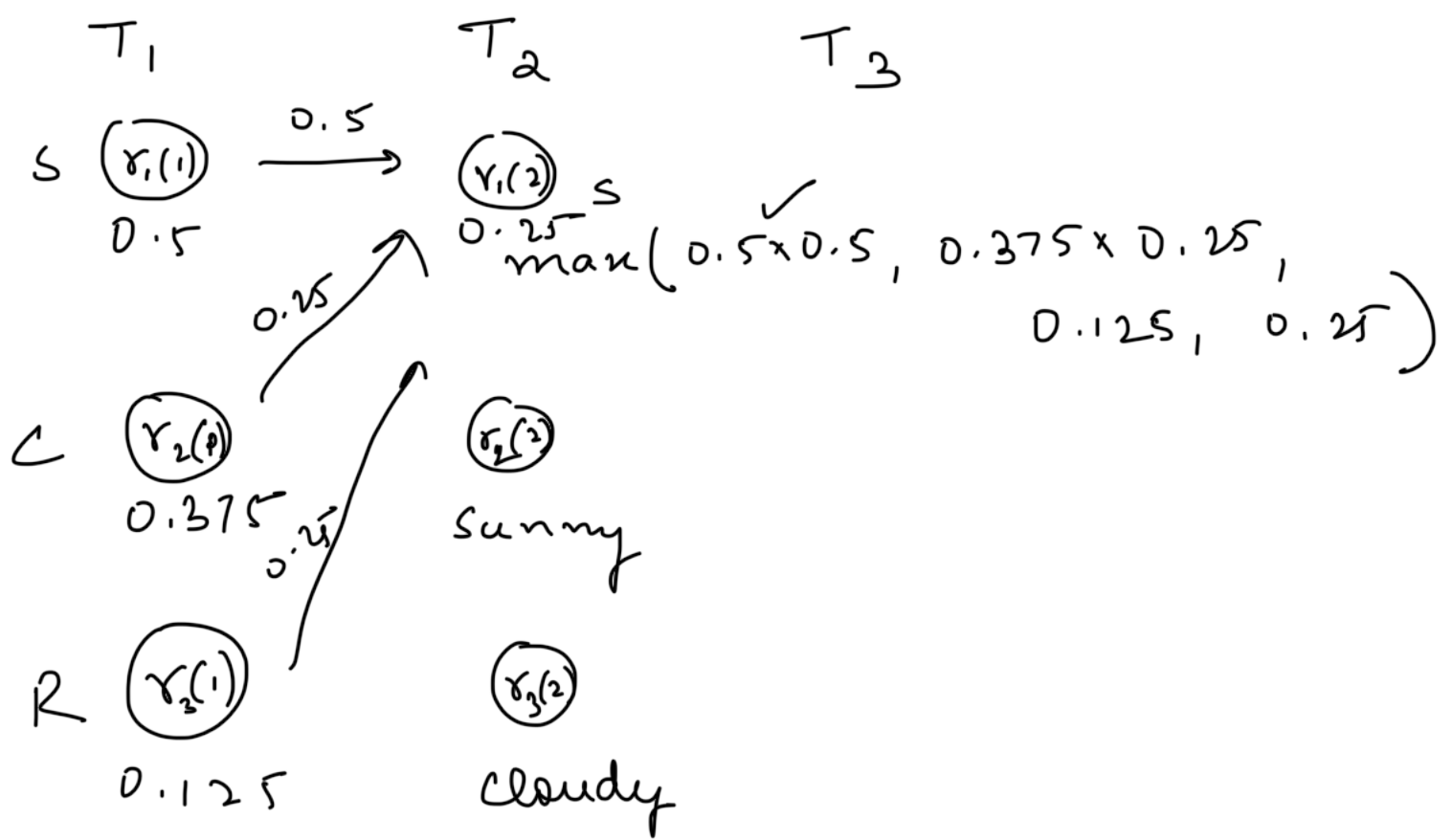
similarly go on.

L6 Viterbi Algo.

- we had transition probs. (first order markov)

N states & T units $\Rightarrow N^T$ computaⁿ
- that's where Viterbi algo. helps.

$$\gamma_3(j) \xrightarrow{a_{31}} \gamma_1(j+1) = \gamma_3(j) \cdot a_{31}$$



day 0 is Sunny,
first row we get
from transiⁿ matrix

	S	C	R
S	0.5	0.375	0.125
C	0.25		
R	0.25		

$S \rightarrow C \rightarrow R$

it saves a lot of computaⁿ.